

COMP4616 – Fundamentals of Machine Learning

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SPRING 2026

WEEK 1: INTRODUCTION



COMP4616 - Fundamentals of Machine Learning

Course materials & Syllabus on Blackboard.

Check Blackboard Announcements & Messages regularly. I will use Blackboard to communicate with you.

Textbook

- Machine Learning, Tom Mitchell McGraw Hill (1997).
- Introduction to machine learning, Ethem Alpaydın. MIT press, 2020.

Workload & Grading

- In-class activities – 30 %
- Project – 10 %
- Midterm exam – 30%
- Final exam – 30%

Makeup Policy

- No makeups for in-class activities.

In case of any attempt of cheating in written examinations or project you will get an **F** for the course grade, and you will be reported to the Dean's office.

Key idea

Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain. Presumably the child brain is something like a notebook as one buys it from the stationer's. Rather little mechanism, and lots of blank sheets.

Alan Turing

What is Learning? What is Machine Learning?

“The activity or process of gaining knowledge or skill by studying, practicing, being taught, or experiencing something.”

Merriam Webster dictionary

“A computer program is said to learn from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, improves with experience **E**.”

Tom Mitchell

A well-defined learning task is given by $\langle P, T, E \rangle$

Examples of Defining the Learning Task

T: Playing checkers

P: Percentage of games won against an arbitrary opponent

E: Playing practice games against itself

T: Recognizing hand-written words

P: Percentage of words correctly classified

E: Database of human-labeled images of handwritten words

T: Driving on four-lane highways using vision sensors

P: Average distance traveled before a human-judged error

E: A sequence of images and steering commands recorded while observing a human driver.

T: Categorize email messages as spam or legitimate.

P: Percentage of email messages correctly classified.

E: Database of emails, some with human-given labels

T: Task
P: Performance
E: Experience

What is Machine Learning?

We try to uncover underlying patterns in data

Machine learning (ML) is concerned with building autonomous agents which uses the experience gained (underlying patterns) from the given data to perform some tasks.

Machine learning and **Artificial Intelligence** are generally confused

AI does not always imply a learning-based system

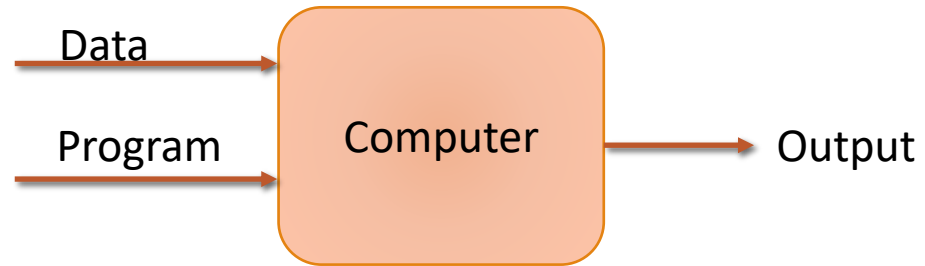
- Symbolic reasoning
- Rule based system
- Tree search
- etc.

Learning based system → learned based on the given data

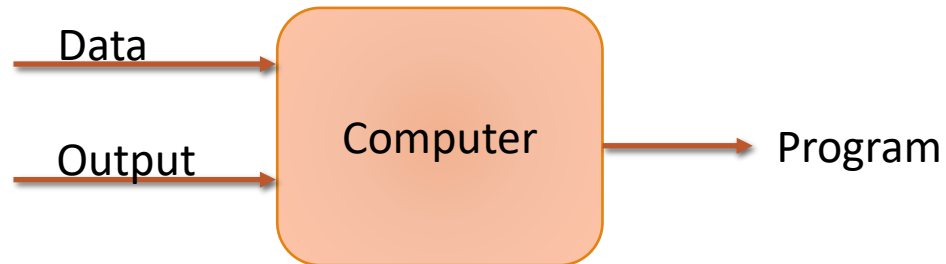
- more flexibility, good at solving pattern recognition problems

Programming vs Machine Learning

Traditional Programming



Machine Learning



When do we use Machine Learning?

For many problems, it's difficult to program the correct behavior by hand

- recognizing people and objects
- understanding human speech
- understanding huge genomic data

Machine learning approach:

- program an algorithm to automatically learn from data, or from experience

Why might you want to use a learning algorithm?

- hard to code up a solution by hand (e.g. vision, speech)
- system needs to adapt to a changing environment (e.g. spam detection, personalized medicine)
- want the system to perform better than the human programmers
- privacy/fairness (e.g. ranking search results)

Don't forget that not all problems need learning.

- No need to learn how to calculate the payroll of employees.

Machine Learning Workflow

1. Should I use ML on this problem?
 - Is there a pattern to detect?
 - Can I solve it analytically?
 - Do I have data?
2. Gather and organize data.
 - Preprocessing, cleaning, visualizing.
3. Choosing a model, loss, regularization, ...
4. Optimization (could be simple or hard).
5. Hyperparameter search.
6. Analyze performance & mistakes, and iterate back to step 3 (or 2)

Types of Machine Learning

Supervised learning

- have labeled examples of the correct behavior

Reinforcement learning

- learning system (agent) interacts with the world and learns to maximize a scalar reward signal

Unsupervised learning

- no labeled examples – instead, looking for “interesting” patterns in the data

Supervised Learning

and	and	and	th	th	th
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and	and	and	th	th	th

Objective is to learn an unknown **target function** f

Input is a **training set** of labelled examples. (x_i, y_i) where $y_i = f(x_i)$

Examples:

- x_i is an image, y_i is giraffe
- x_i is a seismic signal, y_i is explosion

Output is hypothesis h which is close to f (i.e. meaning that h predicts the correct outcome well on unseen examples)

The unseen examples forms the **test set**.

Many possible hypothesis families for h

- Linear models, logistic regression, neural networks, decision trees, nearest-neighbor, etc.

Supervised Learning

Training set examples for different tasks

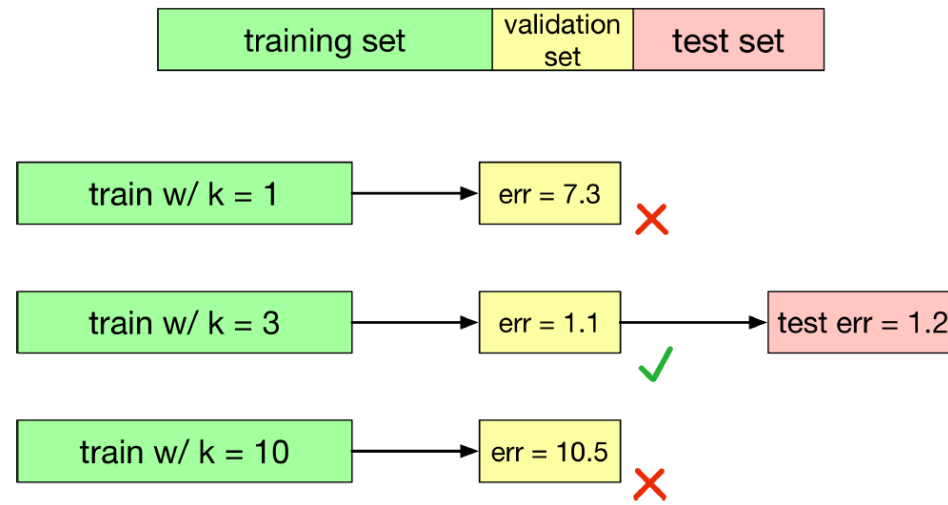
Task	Inputs	Labels
object recognition	image	object category
image captioning	image	caption
document classification	text	document category
speech-to-text	audio waveform	text
⋮	⋮	⋮

Supervised Learning

Most of the times, we would like to **fine tune** the **parameters** in our **hypothesis function**

We can do that by using a separate **validation set**.

Assume that **k** is a parameter that we want to fine tune.



Supervised Learning – Object Recognition

x_i



y_i

giraffe

giraffe

giraffe

llama

llama

llama

$x =$

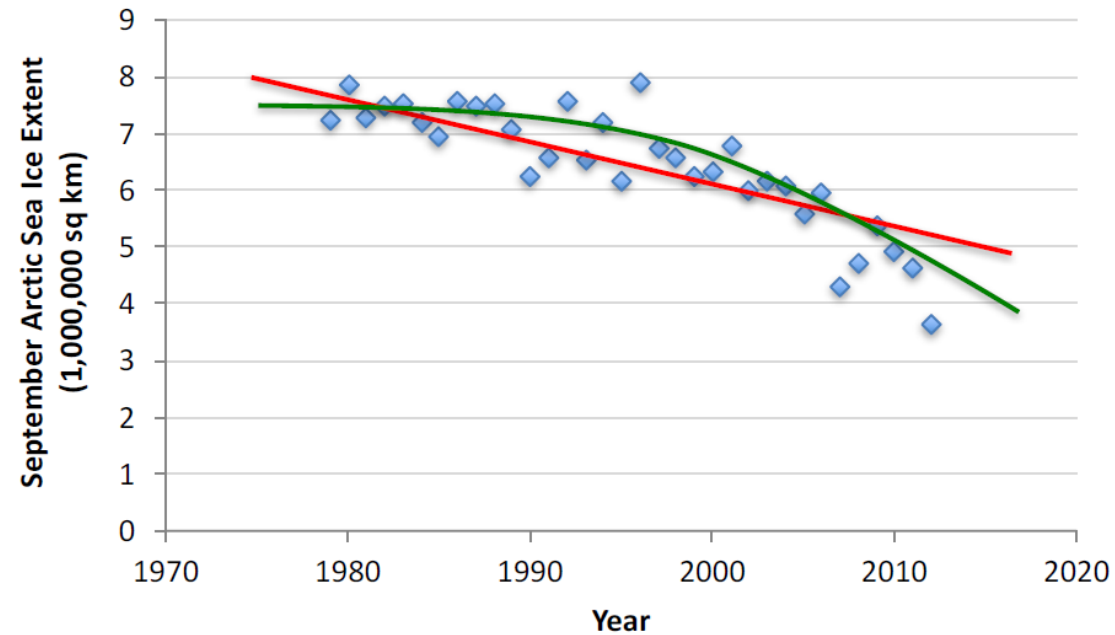


$f(x) = ?$

Supervised Learning

If y is real valued, we call the learning process as **regression**

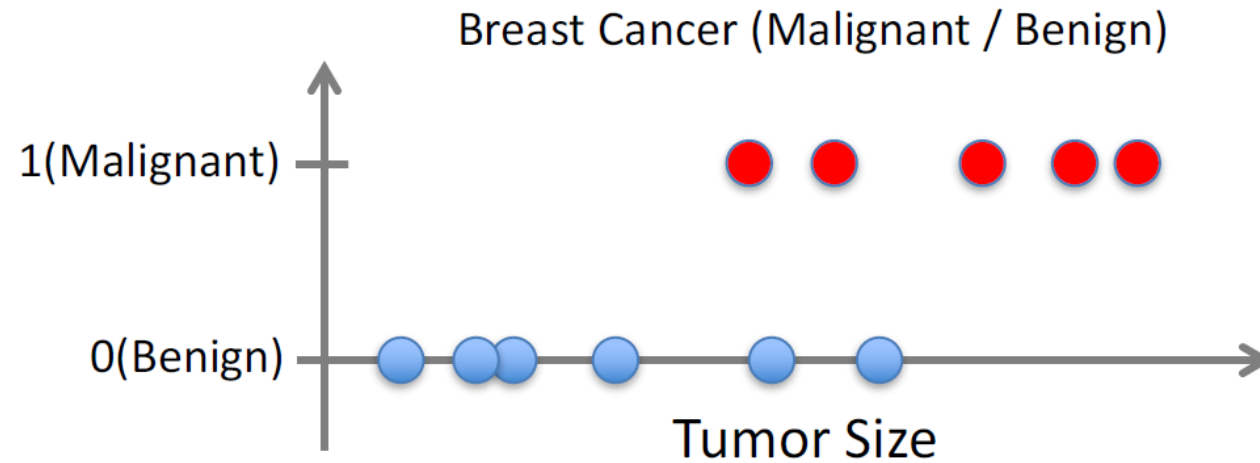
Regression $\rightarrow$ continuous output

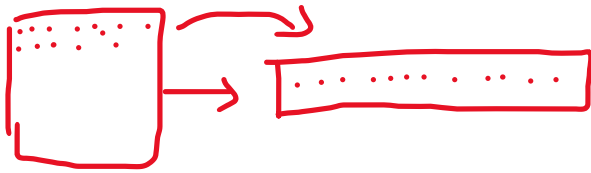


Supervised Learning

If y is categorical, we call the learning process as **classification**

Classification $\rightarrow$ discrete output





Supervised Learning

Which hypothesis space **H** to choose?

- **Example:** Linear combination of features

$$h_w(x) = w^T x$$

How to measure **degree of fit** between the hypothesis h and target function f ?

- Loss function
- **Example:** Squared error

$$\sum_j (y_j - w^T x)^2$$

$$f(x) = y$$

$$h(x) = y'$$

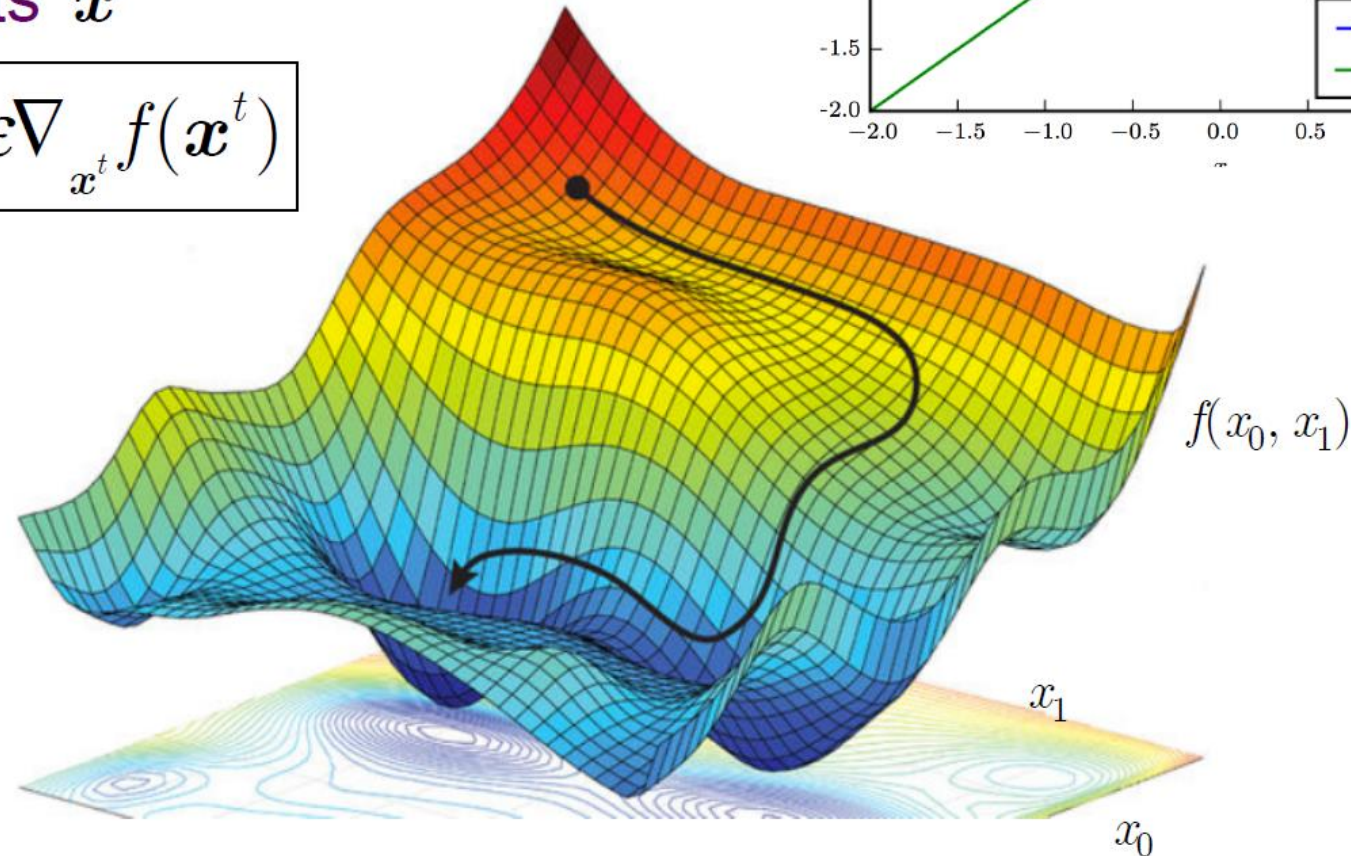
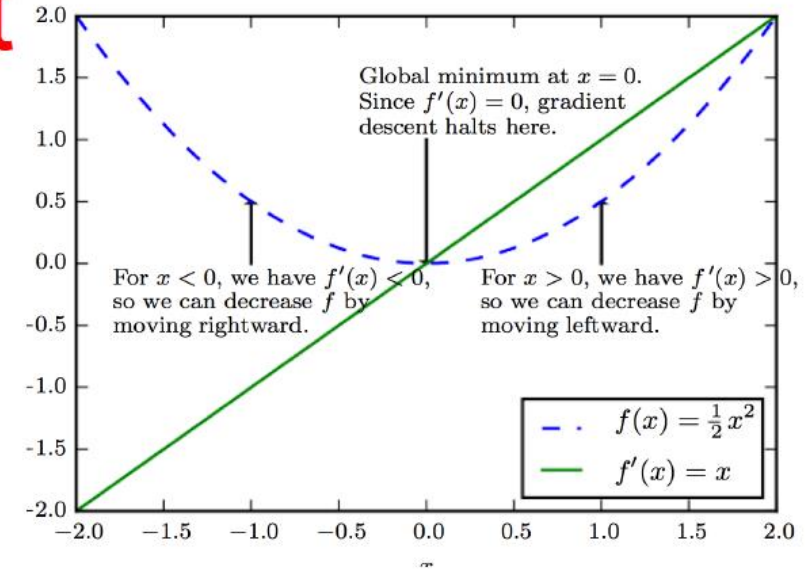
$$h_w(x) = \underbrace{w_1}_{\downarrow} x_1 + \underbrace{w_2}_{\downarrow} x_2 + \underbrace{w_3}_{\downarrow} x_3 + \dots$$

Supervised Learning by Gradient Descent

Classification Task:

Loss function $f(\mathbf{x})$,
e.g., sum of squared errors,
given weights $\mathbf{x}$

$$\mathbf{x}^{t+1} = \mathbf{x}^t - \varepsilon \nabla_{\mathbf{x}^t} f(\mathbf{x}^t)$$

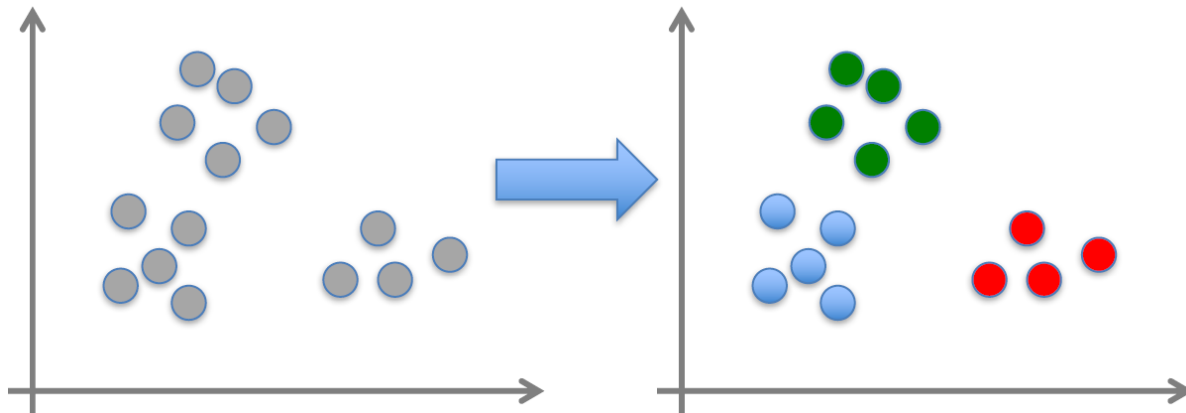


Unsupervised Learning

Given x_i s without labels, we try to output the hidden structure of the data.

Clustering methods

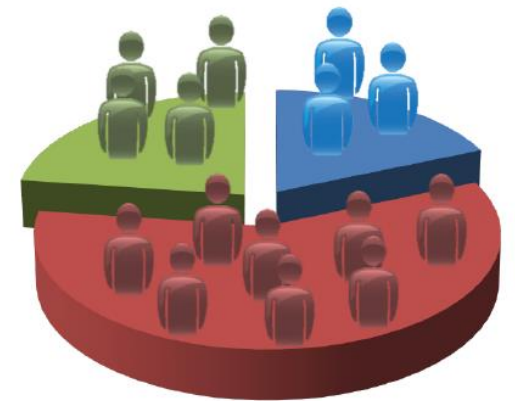
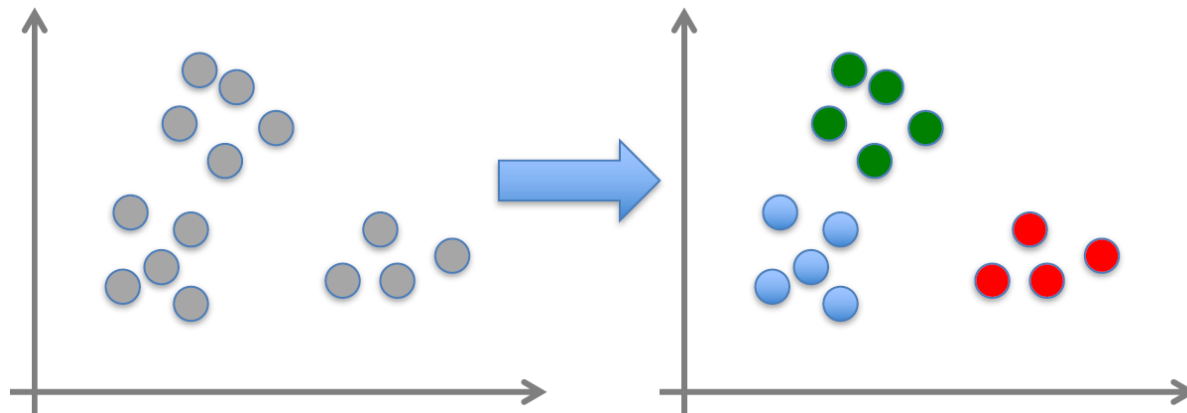
- Used to group similar data points together based on their characteristics.
- They help identify patterns in unlabeled datasets.



Unsupervised Learning Methods

Clustering Methods

Method	Best For	Weakness
K-Means	Large datasets, fast processing	Sensitive to initial choice of k
Hierarchical Clustering	Small datasets, visualization	Slow for large datasets
DBSCAN	Clustering with noise & varying densities	Struggles with varying cluster densities
GMM	Data with overlapping clusters	Computationally expensive



Market segmentation

Reinforcement Learning

Reinforcement Learning (RL) is a type of machine learning that learns through rewards and punishments, much like training a dog.

- Training data is different from supervised/unsupervised learning because:
 - The agent is given feedback (a reward or penalty) instead of explicit labels.
 - The reward signal might be delayed (e.g., winning a chess game only after many moves).

Agent learns by interacting with an environment and receiving rewards for good actions and penalties for bad actions.

Components of RL:

- **Policies:** Define what action to take in a given situation.
- **Utility Estimation:** Measures how good a particular state is.
- **Credit Assignment Problem:** Determines which past actions led to success.

Reinforcement Learning

Applications:

- Game playing (e.g., AlphaGo, AI for Chess, Atari games)
- Robotics (e.g., teaching a robot to walk)
- Autonomous driving

Example Use Case:

- Learning to play Atari games → The AI starts by making random moves, but over time it learns which actions score higher and adapts its strategy.

Reinforcement Learning - Learning to play ATARI

Example: Learning to Play Atari with RL

- The agent sees the game screen as pixel input.
- It chooses an action (e.g., move left, move right, jump).
- The environment responds (e.g., score increases or decreases).
- The agent learns over time which actions maximize long-term rewards.

Q-value, which is the expected cumulative reward for taking action a in state s , and then following the optimal policy afterward

- $\text{Action}(a) = \{\text{left}, \text{right}\}$
- $\text{Observation}(s) = [\text{image frame}]$
- $\text{Reward}(r) = -100$ if lose, -1 if win
- $\text{Policy}(\pi) = P_{\pi}(a|s)$
 - 10,000 states, 2 actions
- $Q(s, a) = \text{value}(\text{action}, \text{state})$
$$Q_{i+1}(s, a) = \mathbb{E}_{s'} [r + \gamma \max_{a'} Q_i(s', a') | s, a]$$
- $\text{Loss} = \gamma + \mathbb{E} [\max_{a'} Q(s', a') - Q_i(s', a')]$

